PROJECT & KEY FINDINGS

Matching payments did not explain every encounter cost

> **Key finding:** Exported charges and payments matched across 9,421 claims, yet eight investigated encounters contained a combined **$596.64 in costs absent from exported charge transactions**.

The project

Assess whether claims and payment reporting was internally consistent, identify records needing investigation, and present the results in a Power BI dashboard. The analysis covered 108 synthetic patients, 5,571 encounters, and 85,047 transaction records.

Results at a glance

Result	Verified finding
Claims analyzed	**9,421**
Exported charges	**$15,795,506.28**
Exported payments	**$15,795,506.28**
Claim-level charge/payment differences above $0.005	**0**
Claims flagged for positive recorded balances	**8**
Encounter costs absent from charges in the eight investigated cases	**$596.64**

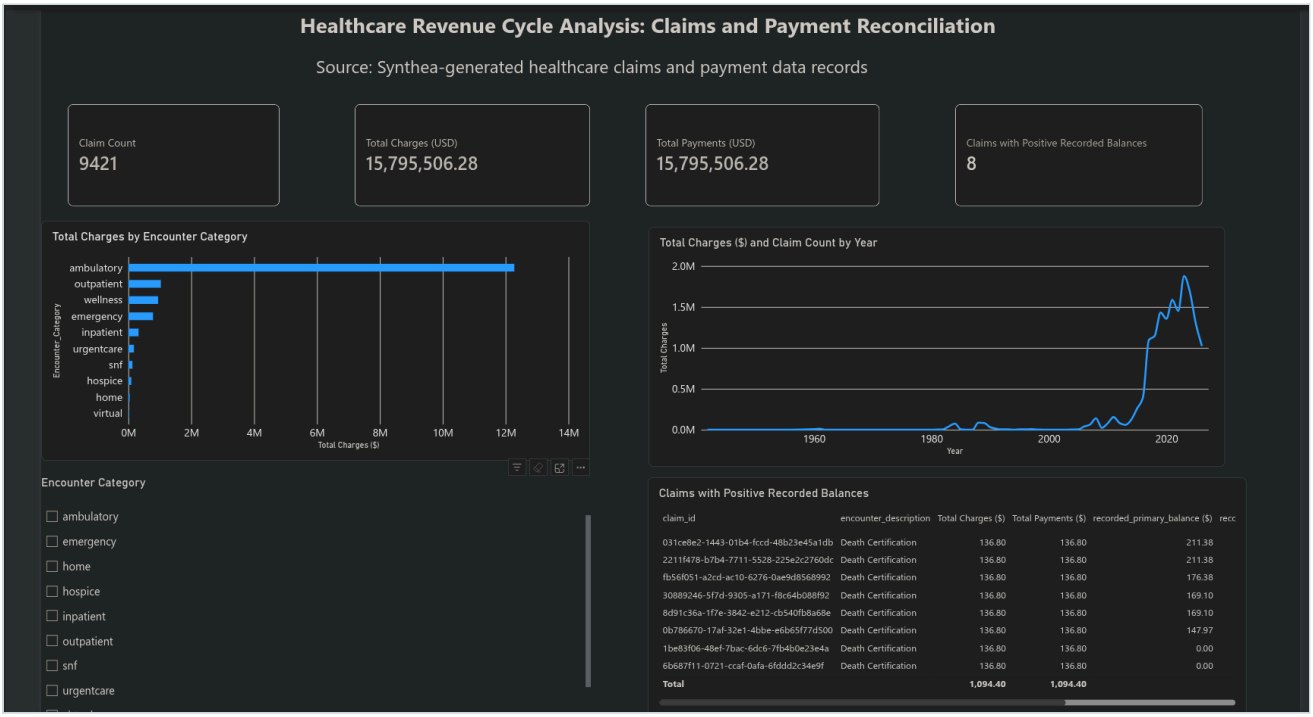


Figure 1. The dashboard brings financial totals, category activity, and flagged claims into one view. Ambulatory encounters account for 77.72% of exported charges.

> **Why it matters:** Matching exported totals are one reconciliation check. They do not establish that the export includes every encounter cost or that recorded balances represent collectible debt.

WORK COMPLETED & RESULTS

From source records to traceable exceptions

Work completed

1. **Profiled and validated source data:** checked identifiers, required financial values, transaction references, and claim-to-encounter/patient consistency.
2. **Prepared claim-level reporting:** aggregated transactions before joining encounter context, preserving all 9,421 claims and avoiding duplicate totals.
3. **Reconciled financial activity:** compared charges and payments overall and per claim using Python and SQL; investigated the eight balance exceptions.
4. **Built and checked the dashboard:** created Power BI measures, category and year visuals, a slicer, and a filtered exception table; reviewed filtering against known totals.

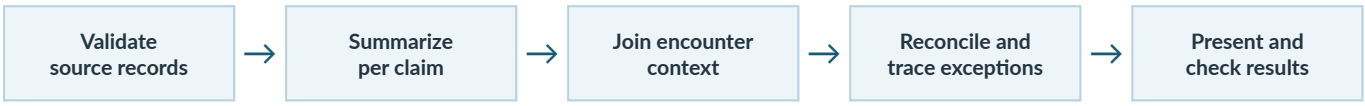


Figure 2. Python and SQL prepare and verify the reporting data; Power BI supports exploration and exception review.

What the exception investigation found

All eight flagged claims were associated with death-certification encounters. Each showed the same comparison:

Per investigated encounter	Amount
Total encounter cost	$211.38
Exported charges	$136.80
Exported payments	$136.80
Encounter cost less exported charges	**$74.58**



Total encounter cost: $211.38 · Same pattern across all eight investigated encounters

$211.38 encounter cost − $136.80 exported charges = $74.58 gap
8 affected encounters × $74.58 = $596.64 combined gap

Reviewed Synthea source code supports an explanation in which report costs enter encounter totals but are excluded from exported item transactions. The archive's exact generator version and configuration are unknown, so this explanation is supported rather than conclusively reproduced for this snapshot.

Verification evidence

- An independent verifier passed **17 checks** against archived CSVs and saved exports.
- Two deliberate changes to copied exports—a balance flag and a charge amount— were both rejected by the verifier.
- Selecting ambulatory showed **4,083 claims**, **$12,277,051.63** in charges and payments each, and **zero flagged claims**. Clearing the selection restored the full totals and eight exception rows.

View the category-filter recording (MP4, approximately 39 seconds) (powerbi/screenshots/slicer-demo.mp4).

IMPACT & RECOMMENDATIONS

Impact, recommendations, and boundaries

Delivered impact

The project identified and quantified a reporting discrepancy that a comparison of exported charges and payments alone would not reveal. It produced reusable validation scripts, an exact-decimal exception trace, and a dashboard for exploring activity and reviewing flagged claims.

These are demonstrated analytical deliverables. Revenue recovery, reduced processing time, and changes in operational performance were not measured.

Recommended actions

Finding	Implication	Recommended action
Charges and payments match, but investigated encounter costs differ	Transaction reconciliation does not establish cost coverage	Check encounter costs separately from exported financial activity
Eight claims have positive recorded balances	A balance flag requires interpretation	Review source allocations and transaction context before treating a balance as debt
All eight investigated gaps share the same encounter type and amount	The pattern supports a focused export investigation	Confirm the generator version and reproduce the behavior before changing export logic
Checks and exception traces are saved	Results can be reviewed beyond the dashboard	Retain source provenance, metric definitions, and validation evidence with reporting outputs

> **Potential business value:** This approach can support more reliable reporting and better-informed exception review. Its effectiveness in a real billing workflow would require validation using that organization's data and rules.

Scope and limitations

The data is synthetic and does not describe an actual provider's performance. The $596.64 is a representation gap, not verified unpaid debt or revenue loss. No generator repair or operational deployment was performed. The independent verifier checked archived data and saved exports; dashboard filtering was reviewed through visual evidence, including the recording. A clean installation and fresh live database reproduction have not been verified.

Tools: Python/pandas · SQL/MySQL · SQLAlchemy · Power BI/DAX.

Supporting evidence: Reproduction guide (REPRODUCIBILITY.md) · Verification results (outputs/verification_summary.json) · Failure controls (outputs/verifier_negative_controls.json) · Exception trace (outputs/balance_exception_trace.csv).

Data source: Synthea sample data.